WORKSHEET

Chapter 7: Integration of reciprocals

1 Find the indefinite integral of each of the following.

a $\dfrac{2}{2x+5}$

b $\dfrac{4x}{2x^2+1}$

c $\dfrac{5x^4}{x^5-2}$

d $\dfrac{1}{2x}$

e $\dfrac{2}{x}$

f $\dfrac{5}{3x}$

g $\dfrac{2x-3}{x^2-3x}$

h $\dfrac{x}{x^2+2}$

i $\dfrac{3x}{x^2+7}$

j $\dfrac{x+1}{x^2+2x-5}$

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2 Find:

a $\int \frac{4}{4x-1}\,dx$

b $\int \frac{dx}{x+3}$

c $\int \frac{x^2}{2x^3-7}\,dx$

d $\int \frac{x^5}{2x^6+5}\,dx$

e $\int \frac{x+3}{x^2+6x+2}\,dx$

f $\int \frac{x+1}{x^2+x+4}\,dx$

3 Evaluate each of the following correct to 1 decimal place.

a $\int_1^3 \frac{2}{2x+5}\,dx$

b $\int_2^5 \frac{dx}{x+1}$

c $\int_1^7 \frac{x^2}{x^3+2}\,dx$

d $\int_0^3 \frac{4x+1}{2x^2+x+1}\,dx$

e $\int_3^4 \frac{x-1}{x^2-2x}\,dx$

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4 Find:

a $\int \frac{6x + 7}{x^2}\, dx$

b $\int \frac{3x^2 + 2x - 2}{x^2} dx$

c $\int \frac{x^2 - 2x - 10}{x^3}\, dx$

d $\int \frac{2x^4 - x + 12}{x^3}\, dx$

e $\int \frac{x^2 + 5x - 3}{x - 2}\, dx$

f $\int \frac{2x^2 - 5x - 2}{x - 4}\, dx$

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Answers

1 **a** $\log_e(2x+5)+c$

b $\log_e(2x^2+1)+c$

c $\log_e(x^5-2)+c$

d $\frac{1}{2}\log_e(x)+c$ or $\frac{1}{2}\log_e(2x)+c$

e $2\log_e(x)+c$

f $\frac{5}{3}\log_e(x)+c$

g $\log_e(x^2-3x)+c$

h $\frac{1}{2}\log_e(x^2+2)+c$

i $\frac{3}{2}\log_e(x^2+7)+c$

j $\frac{1}{2}\log_e(x^2+2x-5)+c$

2 **a** $\log_e(4x-1)+c$

b $\log_e(x+3)+c$

c $\frac{1}{6}\log_e(2x^3-7)+c$

d $\frac{1}{12}\log_e(2x^6+5)+c$

e $\frac{1}{2}\log_e(x^2+6x+2)+c$

f $\frac{1}{2}\log_e(x^2+2x+4)+c$

3 **a** 0.5

b 0.7

c 1.6

d 3.1

e 0.5

4 **a** $6\log_e(x)-\frac{7}{x}+c$

b $3x+2\log_e(x)+\frac{2}{x}+c$

c $\log_e(x)+\frac{2}{x}+\frac{5}{x^2}+c$

d $x^2+\frac{1}{x}-\frac{6}{x^2}+c$

e $11\log_e(x-2)+\frac{x^2}{2}+7x+c$

f $10\log_e(x-4)+x^2+3x+c$

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